AquaGrid

An app that works out the smartest time to make drinking water on a small island — so the island wastes less fuel and never runs dry.

Rishik · Neekin · Srihan · Ednit · Gowtham

DreamHacks 2026 · Track 2 — AI, Automation & Logic

Small islands have to make their water

Rain fills the tanks. When the rain stops, a machine turns seawater into drinking water.

That machine is the **hungriest thing on the island** for electricity.

By day that power is free — it's sunshine. At night it's a diesel generator, and the diesel comes **by boat every few weeks**.

Midday

sunshine — free

2 a.m.

diesel — costs money

The tank

holds about 2 days

The tank is basically a **battery for water**. You can fill it whenever you like — so you should fill it when power is free.

But nobody chooses. A timer does.

Most islands run the machine on a **timer set to the middle of the night**, because that's when electricity is cheap on the mainland.

On a solar island that's backwards — the middle of the night is exactly when there is **no sun**.

This is a real problem. In **2011 Tuvalu ran down to a few days of drinking water** and had to declare a state of emergency.

Their machine wasn't too small. The problem was that **nobody could see what was coming**.

So we taught it to see what's coming

01	02	03	04
Predict	**Decide**	**Explain**	**Detect**
How much water the island will use, 2 days ahead	The best hours to run the machine	Says why, in plain English	Spots problems early

We trained an AI on **2 years of water use**. Then we tested it on days it had **never seen before** — real future days, not a shuffled deck.

45 litres

how far off it is, per hour — out of about 940

0.987

accuracy score, where 1.0 is perfect

89%

better than just guessing the average

It uses a fifth of the fuel

HOW IT'S RUN	DIESEL USED	COST	FUEL LASTS	RESULT
AquaGrid	**15 L**	**$25**	**39 days**	**fine**
Night timer	70 L	$112	8.6 days	runs out
Fill when low	79 L	$127	7.6 days	runs out

Here's the part that really matters. The island has **300 litres of diesel** and the boat is **12 days away**.

The other two ways **run out of fuel before the boat arrives**. Ours doesn't.

The whole thing runs on the phone

THE APP	**React Native**	what you see and tap — we drew every chart ourselves
THE BRAIN	**RandomForest AI**	we trained it on a laptop, then packed it **inside the app**
THE WEATHER	**Open-Meteo**	the phone grabs its own sun and rain forecast — free, no sign-up
THE VOICE	**Groq · GPT-OSS 120B**	an open AI model that explains things in plain English

There is **no server anywhere**. An app that needs the internet to decide when to make water would stop working **exactly when an island gets cut off** — which is when you need it most.

We put the whole AI inside the app

Normally an AI lives on a big computer and your phone asks it questions over the internet. We did the maths **ahead of time** for every situation the AI could ever face, and packed all the answers into the app.

15,120
answers — the whole AI, not a sample

0.011 ms
time to predict a whole day, on the phone

Any island
works for places the laptop never saw

The RandomForest

We trained this. It makes

every

decision.

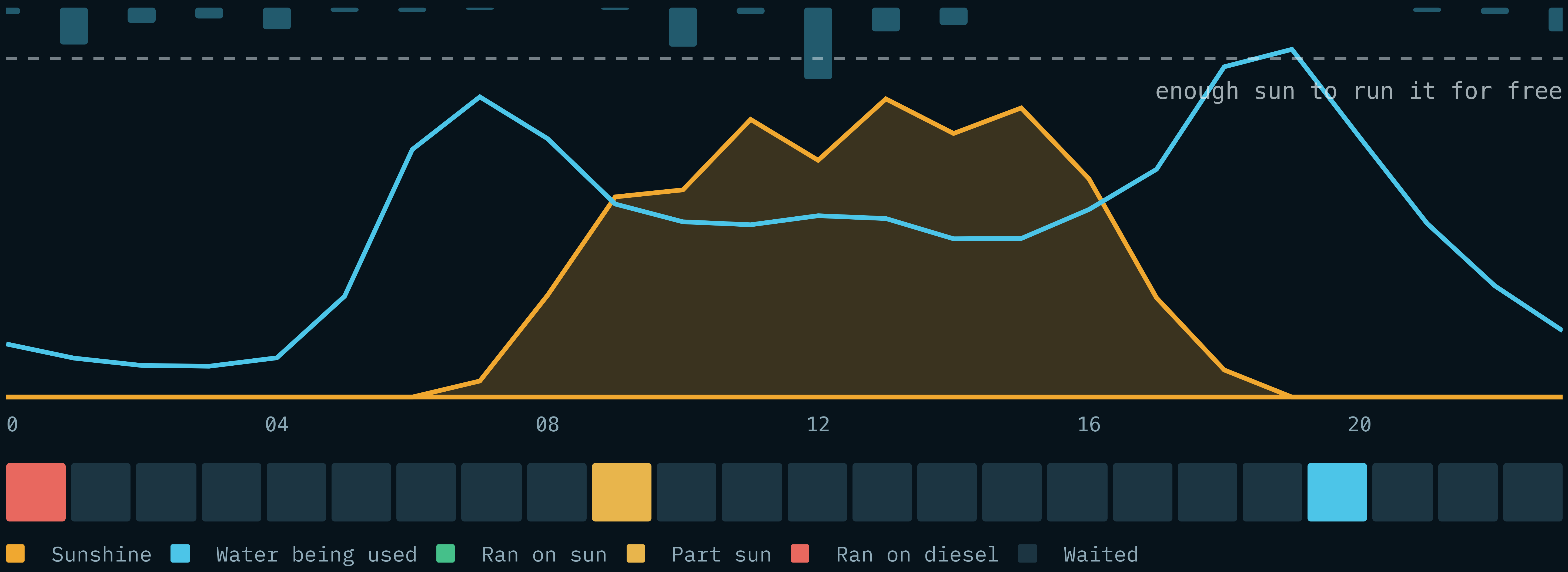
The language model

Only writes the words. Delete it and

every number stays the same

.

One real day



Orange above the dotted line means the sun alone can run the machine. The strip underneath is what it actually chose to do, hour by hour.

It also catches problems early

It knows how fast the tank should empty. So when the real tank drops faster than that, something is wrong — and it says so.

Burst pipe
found in 2 hours

Dirty solar panels
losing 13% of the sunshine

Worn-out filters
need cleaning in 40 days

Running low on fuel
before the boat comes

It saves the rain too. Rain off the rooftops gave **24%** of all the water — so if rain is coming, it waits instead of wasting fuel.

And it stores sunshine. A battery carries midday sun into the evening, when everyone gets home and uses the most water.

Let’s show you

1 What it’s doing right now — and **why**, in a full sentence	2 The day ahead: green = free sun, blue = stored sun, red = diesel
3 How much fuel it saves	4 **Press “burst pipe”** — watch the alarm appear

Nobody would have spotted that leak for hours. The app found it in two.

And then

5 **Press "cyclone"** — it stops saving fuel and fills the tank

6 Switch to a different island — it works everything out again

7 Ask it a question. Then ask "why?"

8 **Turn on aeroplane mode** — everything still works

Because on a real island, the internet is the first thing to go.

How it's built

755

lines of code make all the decisions

One command

tests it all — no phone needed

Free

the weather data costs nothing and needs no sign-up

ml/	**Teaches the AI**	2 years of water use → a trained model
logic/	**Makes the decisions**	when to run, and spotting problems
lib/	**Talks to the world**	weather, location, the language model
components/	**Draws the screen**	18 pieces, every chart hand-drawn

Everything is written down

Website
the charts are made from real data, not screenshots

35 pages
a handbook explaining every file

All open
anyone can read it and check our numbers

We didn’t just say it saves fuel. **We ran three different ways of doing it** — ours, the night timer, and fill-when-low — on the same weather, and compared them.

github.com/sbongula/aquagrid

Every number here is one we measured.

We tested the AI on days it had never seen. We gave the leak detector a real leak and made it find it. And when we tried Reykjavík — where there's barely any sun — it only saved 23%. We left that in, because a smaller honest number is worth more than a big one we can't back up.

github.com/sbongula/aquagrid

sbongula.github.io/aquagrid